

MANCHIKANTI SURYA VARDHAN

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Professional Summary

M.Tech VLSI Design student with hands-on experience in RTL design, FPGA prototyping, ASIC implementation, and custom IP core development. Proficient in Verilog HDL, Cadence Genus, Innovus, NCLaunch, Xilinx Vivado, Vitis AI, and Vitis HLS. Experienced in RTL-to-GDSII flow, logic synthesis, physical design, timing closure, FPGA-based AI acceleration, and functional verification. Strong understanding of Digital IC Design, CMOS VLSI Design, FPGA Architecture, and RISC-V Processor Design. Seeking opportunities in ASIC Design, FPGA Design, Physical Design, SoC Development, and EDA/CAD Engineering.

Technical Skills

HDL / Languages	Verilog HDL, SystemVerilog (learning), TCL, Linux Shell Scripting
EDA Tools	Cadence Genus, Cadence Innovus, Cadence NCLaunch, QuestaSim, ModelSim, Vivado, Vitis AI, Vitis HLS, Quartus Prime
ASIC Flow	RTL-to-GDSII Flow, Logic Synthesis, Floorplanning, Placement, CTS, Routing, STA
FPGA / SoC	Xilinx ZCU104 MPSoC, IP Integrator, DPU Integration, FPGA Prototyping, Altera DE10
Verification	Functional Simulation, Testbench Development, Waveform Debugging, SystemVerilog Assertions (SVA Basics)
Architecture	RISC-V (RV32I), MAC Unit Design, FIR Filter Design, SoC/IP Integration
Core Concepts	Digital Logic Design, CMOS VLSI Design, Digital IC Design, FPGA Architecture, ASIC Design Flow, Physical Design

Projects

32-bit RISC-V Processor with Integrated MAC Unit — RTL-to-GDSII ASIC Flow

- Designed and implemented a 32-bit single-cycle RISC-V processor with an integrated MAC Unit using Verilog HDL.
- Executed complete ASIC Physical Design flow including Floorplanning, Power Planning, Placement, Clock Tree Synthesis (CTS), Routing and Sign-off Checks.
- Performed logic synthesis using Cadence Genus and physical implementation using Cadence Innovus with gpdk90 (90nm) technology library.
- Applied SDC timing constraints and analyzed setup and hold timing violations throughout implementation stages.
- Performed timing optimization across placement, CTS and post-route stages to achieve timing closure.
- Generated utilization, congestion, timing and power reports for design quality assessment.
- Verified post-route design integrity through DRC checks and physical verification reports.

FPGA-Based Drone Detection and Classification System using YOLOv8n

- Implemented a hardware-accelerated drone detection pipeline on FPGA using lightweight **YOLOv8n** architecture for real-time edge AI inference.
- Trained and fine-tuned the YOLOv8n model on a custom drone dataset, improving detection precision by approximately **5.3%**.
- Exported the trained model to **ONNX** format, performed INT8 quantization using Vitis AI, and deployed it on the **Xilinx ZCU104 MPSoC** using DPU acceleration.
- Designed and integrated a custom IP Core for Non-Maximum Suppression (NMS) using Vivado HLS, enabling hardware-accelerated post-processing of object detections.

VLSI Design of 2D FIR Filter Architecture using CSD

- Designed a 2D FIR Filter Architecture using **Canonical Signed Digit (CSD)** representation to eliminate conventional multipliers.
- Implemented multiplier-less DSP architecture using optimized shift-and-add operations, reducing hardware complexity and power consumption.
- Generated and optimized FIR filter coefficients using MATLAB and implemented the architecture using Verilog HDL in Xilinx Vivado.
- Achieved improved area and power efficiency compared to traditional multiplier-based FIR filter implemen-

tations.

Education

Vellore Institute of Technology, Chennai Master of Technology (M.Tech) in VLSI Design CGPA: 8.31	2025 – Present
CVR College of Engineering, Hyderabad Bachelor of Technology (B.Tech) in Electronics and Communication Engineering CGPA: 8.23	2019 – 2023
Narayana Junior College, Hyderabad Intermediate (MPC) Percentage: 89.9%	2017 – 2019
Narayana E-Techno School, Hyderabad SSC (10th Grade) CGPA: 9.3	2017

Relevant Coursework

- Digital IC Design
- CMOS VLSI Design
- ASIC Design Flow
- FPGA Architecture
- Static Timing Analysis
- Physical Design
- Verification Methodologies

Certifications

- **Semiconductor Skill Accelerator (SSA) Program** — IIT Bombay & VLSI System Design (VSD) 2026
- **Workshop on RTL-to-GDSII Physical Design Flow** — ProV Logic Jun 2026
- **Workshop on VLSI Design & Semiconductor Industry** — ChipXpert Technologies Pvt. Ltd. May 2026
- **ASIC Flow, Logic Synthesis, STA, Physical Design** — Cadence Genus, Tempus, Innovus & Synopsys ICC2 (Udemy - VLSI Mentor) Ongoing
- **Certified in CMOS Digital VLSI Design** — NPTEL / SWAYAM 2025
- **Certified in Verilog HDL — Hands-On** — Maven Silicon 2025